SIMATS ENGINEERING

Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Max Marks:100

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ANNEXURE A

ANALYTICAL PROBLEM SOLVING

Q1. Negative Transformation – X-ray Component Inspection

An X-ray inspection system examines packaged electronic components. Six grayscale pixels representing regions of different material densities have intensities:

r = [18, 42, 76, 134, 188, 236]

For an 8-bit image, apply s = 255 − r.

Tasks:

a) Calculate the negative-transformed intensity for every input pixel.
b) Calculate the mean and intensity range of the original and transformed pixel sets.
c) Determine the original and transformed intensity difference between r = 42 and r = 188. Explain whether their absolute contrast changes.
d) Identify which original pixels become brighter than 200 after transformation.
e) Using your calculations, evaluate how inversion changes the visibility of low-intensity structures without increasing their intensity separation.

Q2. Contrast Stretching – Fog-Affected Road Image

A roadside monitoring camera records a low-contrast image during light fog. Sample pixel intensities from the roadway and nearby objects are:

r = [72, 84, 96, 108, 120, 132, 144]

Map rmin = 72 and rmax = 144 to the output interval 0–255 using:

s = [(r − rmin) / (rmax − rmin)] × 255. Round outputs to two decimal places.

Tasks:

a) Calculate the stretched intensity for each input pixel.
b) Calculate the original and output dynamic ranges and the ratio of the output range to the original range.
c) Find the intensity difference between pixels r = 96 and r = 120 before and after stretching.
d) If an object is separated from the road surface by 12 original grayscale levels, calculate the corresponding separation after transformation.
e) Based on these calculations, assess the benefit and limitation of linear contrast stretching in a fog-affected image.

Q3. Logarithmic Transformation – Astronomy Image Analysis

A telescope image contains weak stars and a bright celestial object. The selected pixel intensities are:

r = [0, 2, 6, 15, 35, 75, 150, 250]

Apply s = 100 log10(1 + r). Round outputs to two decimal places.

Tasks:

a) Calculate the transformed intensity of every pixel.
b) Calculate the output difference for the low-intensity pair r = 2 and r = 6 and the high-intensity pair r = 150 and r = 250.
c) Calculate the corresponding input differences and the output-to-input difference ratios for both pairs.
d) Determine the transformed dynamic range of the listed pixels and whether any calculated outputs exceed 255.
e) Interpret whether the logarithmic mapping preferentially expands dark-region differences or bright-region differences, using your calculated ratios.

Q4. Gamma Transformation – Uneven Illumination in Document Scanning

A document scanner captures normalized pixel intensities from faint handwriting and the page background:

r = [0.04, 0.16, 0.36, 0.64, 0.81, 1.00]

Use s = r^γ for γ = 0.4 and γ = 1.8. Round outputs to three decimal places.

Tasks:

a) Calculate all six output values for each gamma setting.
b) For both gamma settings, calculate the output difference between r = 0.04 and r = 0.16.
c) Calculate the mean normalized intensity of the original set and of each transformed set.
d) Identify which setting raises the intensities of darker pixels, and quantify the increase at r = 0.16.
e) Based on the results, recommend a gamma setting for making faint dark-region handwriting more visible, while noting a possible drawback.

Q5. Histogram Equalization – Low-Contrast Microscopy Image

A 3-bit microscopy image has 40 pixels distributed over eight intensity levels as follows:

Intensity level rk: 0 1 2 3 4 5 6 7

Pixel frequency nk: 0 3 5 12 10 6 3 1

Use L = 8, p(rk) = nk / 40 and sk = round[(L − 1) × CDF(rk)] (nearest integer).

Tasks:

a) Calculate the probability distribution for all eight intensity levels.

b) Calculate the cumulative distribution function (CDF) for each level.

c) Determine the equalized intensity value for each original level using the stated rounding rule.

d) Construct the mapping table and determine the output histogram frequencies after combining levels mapped to the same output.

e) Compare the occupied intensity ranges before and after equalization, and discuss whether a wider occupied range guarantees a perfectly uniform histogram.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 AT1 Analytical problem solving

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Course : computer vision

1. a) negative transformed intensity :

Formula S = 255 - r

255 - 18 = 237

255 - 42 = 213

255 - 76 = 179

255 - 134 = 121

255 - 188 = 67

255 - 236 = 19

Transformed pixels : S = [237, 213, 179, 121, 67, 19]

b) mean and intensity range

Original pixels :

Sum = 18 + 42 + 76 + 134 + 188 + 236 = 694

mean = 694/6 = 115.67

Range = 236 - 18 = 218

Transformed pixels :

Sum = 237 + 213 + 179 + 121 + 67 + 19 = 836

mean = 836/6 = 139.33

Range = 237 - 19 = 218

c) intensity difference and absolute contrast

Original Intensity $|188 - 42| = 146$

Transformed intensity $|213 - 67| = 146$

d) pixels that become brighter than 200

The transformed intensity greater than 200 the corresponding original pixels 18 and 42

e) Effect inversion visibility.

Negative transformation converts low intensity pixels into high intensity pixels low intensity pixels. original low intensity pixels 18 and 42 beome 237 and 213. It contrast difference remains unchanged distance between intensity values.

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2. a) Stretched intensity Calculation.

Formula: $S = \frac{r - r_{min}}{r_{max} - r_{min}} \times 255$

$r_{min} = 72$, $r_{max} = 144$ $S = \frac{r-72}{144-72} \times 255 = \frac{r-72}{72} \times 255$

$r = 72$: $S = 0.00$

$r = 84$: $S = 42.50$

$r = 96$: $S = 85.00$

$r = 108$: $S = 127.50$

$r = 120$: $S = 170.00$

$r = 132$: $S = 212.50$

$r = 144$: $S = 255.00$

output [0, 42.50, 85.00, 127.50, 170.00, 212.50, 255.00]

b) Dynamic ranges and ratio

original dynamic range: $144 - 72 = 75$

output dynamic range: $255 - 0 = 255$

$\frac{255}{72} = 3.54$

C) Intensity difference before after stretching.

original difference $r=96$ and $r=120$

$120-96=24$

output difference $170-85=85$

d) Seperation of object

original = 12 grayscale

$$\frac{255}{144-72} = \frac{255}{72}$$

$$12 \times \frac{255}{72} = 42.50$$

e) Benefit and limitation

Benefits: Improves image contrast and makes road surface and object more visible.

Limitation: cannot remove fog or recover lost details may amplify noise.

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3. Logarithmic Transformation

a) Formula $S=100\log_{10}(1+r)$

$r=0$: $S=0.00$

$r=2$: $S=47.71$

$r=6$: $S=84.51$

$r=15$: $S=120.41$

$r=35$: $S=155.67$

$r=75$: $S=188.28$

$r=150$: $S=218.12$

$r=250$: $S=240.00$

output = [0.00, 47.71, 84.51, 120.41, 155.67, 188.28, 218.12, 240.00]

b) output Intensity difference

Low intensity pair (2 and 6)

$84.51 - 47.71 = 36.80$

High intensity pair (150 and 250]

$240.00 - 218.12 = 21.88$

c) Input differences and ratios

Low intensity = 6 - 2 = 4 Ratio = 36.80/4 = 9.20

High intensity = 250 - 150 = 100 = 21.88/100 = 0.22

d) Transformed dynamic range

240.00 - 0.00 = 240.00

e) interpretation:

Logarithmic transformation preferentially dark region difference the low intensity (9.20) higher (0.22) compressing difference bright objects.

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4. Gamma Transformation:

a) output values

$\gamma = 0.4$

$0.04^{0.4} = 0.276$

$0.16^{0.4} = 0.480$

$0.36^{0.4} = 0.665$

$0.64^{0.4} = 0.833$

$0.81^{0.4} = 0.919$

$1.00^{0.4} = 1.00$

output [0.276, 0.480, 0.665, 0.833, 0.919, 1.000]

$\gamma = 1.8$

$0.04^{1.8} = 0.003$

$0.16^{1.8} = 0.037$

$0.36^{1.8} = 0.167$

$0.64^{1.8} = 0.445$

$0.81^{1.8} = 0.684$

$1.00^{1.8} = 1.000$

output: [0.003, 0.037, 0.167, 0.445, 0.684, 1.000]

b) output differences

$\gamma = 0.4$ 0.480 - 0.276 = 0.204

$\gamma = 1.8$ 0.037 - 0.003 = 0.034

c) mean normalized intensity

$\frac{3.01}{6} = 0.502$

$\gamma = 0.4$ output 0.696

$\gamma = 1.8$ output = 0.389

d) Dark pixel intensity

$\gamma = 0.4$

$r = 0.16$ $\quad 0.480 - 0.160 = 0.320$

e) Recommendation

$\gamma = 0.4$ faint more visible brightening dark regions. A possible drawback bright background appear washed out.

5.

a) probability distribution

$$P(r_k) = n_k / 40$$

$P(0) = 0/40 = 0.000$ $\quad$ $P(4) = 10/40 = 0.250$

$P(1) = 3/40 = 0.075$ $\quad$ $P(5) = 6/40 = 0.150$

$P(2) = 5/40 = 0.125$ $\quad$ $P(6) = 3/40 = 0.075$

$P(3) = 12/40 = 0.300$ $\quad$ $P(7) = 1/40 = 0.025$

b) cumulative distribution function (CDF)

CDF values: 0, 0.075, 0.200, 0.500, 0.750, 0.900, 0.975, 1.000

c) Equalized intensity:

Formula $S_k = \text{round}\ [7 \times CDF(r_k)]$

output values: 0, 1, 1, 4, 5, 6, 7, 7

d) mapping and output histogram

mapping $0 \to 0$, $1 \to 1$, $2 \to 1$, $3 \to 4$, $4 \to 5$, $5 \to 6$, $6 \to 7$, $7 \to 7$

output frequencies:

Level 0: 0 pixels

Level 1 : 8 pixels

Level 2 : 0 pixels

Level 3 : 0 pixels

Level 4 : 12 pixels

Level 5 : 10 pixels

Level 6 : 6 pixels

Level 7 : 4 pixels

e) comparison:

original occupied range = 1-7 Equalized occupied range 1-7

Equalization redistributes intensities but the histogram not perfectly uniform because multiple input levels map to same output level.